Universität Zürich[UZH]

Center for Reproducible Science and Research Synthesis

Quantifying reproducibility - Results from a scoping review on reproducibility metrics and simulation studies into their real-world applicability

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Part of improving Reproducibility In ScienceE (iRISE)

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What do we mean by “Reproducibility”?

Definition of reproducibility, for the purpose of this presentation:

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- *Computational reproducibility*:
  Using the *same data and code* to reproduce the original results.
- *Results reproducibility*:
  New data is collected and analysed using similar methods and materials.
- *Partial reproducibility*:
  Only parts of the study are reproduced (e.g., experiment, data collection, analysis), and results are compared.

The million-dollar questions

— *When do results agree?*

— *What does it mean for study results to be consistent?*

Reproducibility Metrics Scoping Review | Question

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Research questions:

- Which metrics have been used or suggested to quantify or assess which type of reproducibility?
- Which ones were suggested only, or have a clear implementation?
- How can they be interpreted and in which situations are they useful?
- What are assumptions and limitations?

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Reproducibility Metrics Scoping Review | Methods

Scoping Review in two parts

— Application papers – For reproducibility metrics **used in practice**, a list of large-scale reproducibility projects was compiled by the project team.

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> **Large-scale reproducibility projects**
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> Large-scale efforts to measure the reproducibility of a field, method or type of study.

— Methodological papers – For reproducibility metrics **proposed, suggested or discussed**, a systematic search was conducted to identify relevant literature.

Application papers

Application papers

- **49 records** of reproducibility projects across fields of research (based on OpenAlex):
  - Health and life sciences – 10 (20.4%)
  - Physical sciences – 6 (12.2%)
  - Social sciences – 33 (67.3%)

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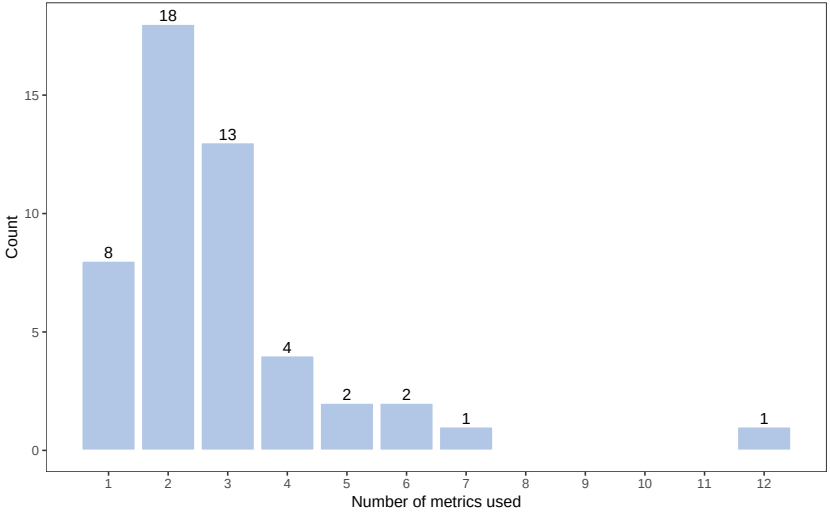
Type of Metric	N (%)
Agreement in statistical significance	32 (65.3%)
Agreement in effect size	35 (71.4%)
Meta-analysis of study results	9 (18.4%)
Subjective assessment	14 (28.6%)

Number of metrics used in application papers



Methodological Papers | Types of Metrics

— **50 reproducibility metrics** identified across **97 records**.

Methodological Papers | Types of Metrics

- **50 reproducibility metrics** identified across **97 records**.
- Metrics varied by type:

Metric Type	N (%)	Example(s)
Formula or statistical model	37 (74%)	Significance criterion, Bayes factor, Meta-analysis
Framework (author-defined)	3 (6%)	Causal replication
Graph	3 (6%)	Bland–Altman Plot
Study / survey / questionnaire	4 (8%)	Prediction markets, RepliCats
Algorithm	3 (6%)	Various automated assessments

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Review

Check for updates

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A scoping review on metrics to quantify reproducibility: a multitude of questions leads to a multitude of metrics

Rachel Heyard[1], Samuel Pawel[2], Joris Frese[3], Bernhard Voelkl[4], Hanno Würbel[4], Sarah McCann[5], Leonhard Held[1,2], Kimberley E. Wever[6], Helena Hartmann[7], Louise Townsin[8] and Stephanie Zellers[9]

Full paper:
https://doi.org/10.1098/rsos.242076

Best metrics?

It depends! on context, on your question, on your data, on your resources, etc…

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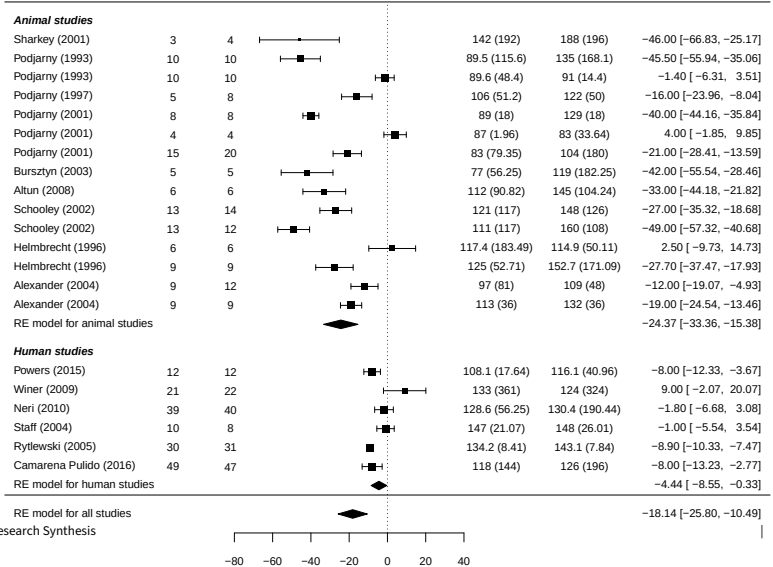
Special Case: Translation of findings from animal models to clinical trials

Animal and human studies effect sizes and effect size variations differ

Data from:
https://doi.org/10.3390/nu12092535



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- There is no single *best* reproducibility metric *(unsurprisingly)*
- Each metric captures its *own aspect* of reproducibility and we linked each metric to a specific research question
- Metrics require different types of input data and some metrics are *costly* to implement
- Ongoing work on the applicability of metrics in practice, including a simulation study to assess the real-world applicability of metrics – *stay tuned*

Thank You!

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